Empirical Results Report

Miratrix

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1 Introduction

This report generates the results reported in our paper (but not the figures—see the script `final_paper_figures.R` for those).

2 Some statistics on the studies

Our count of studies is the following:

	Blocked	Unblocked
Curated	11	0
Public	15	23

Our count of outcomes is the following:

	Blocked	Unblocked
Curated	28	0
Public	22	38

Some statistics about the blocked studies across the 50 outcomes they entail:

statistic	mean	median	sd	min	Q10	Q90	max
J	483.74	56.50	1637.23	16.00	30.00	495.00	8375.00
Jcv	0.40	0.33	0.30	0.00	0.09	0.85	1.00
K	35.16	16.00	65.39	5.00	5.00	85.30	389.00
R2.1	0.21	0.17	0.17	0.00	0.02	0.44	0.58
R2.2	0.43	0.37	0.32	0.00	0.06	0.83	0.95
R_ate	0.06	0.04	0.06	0.00	0.01	0.14	0.30
R_se	1.33	1.25	0.38	1.04	1.06	1.63	3.46
block_ICC	0.09	0.07	0.10	0.00	0.00	0.21	0.39
cluster_ICC	0.09	0.09	0.06	0.00	0.01	0.15	0.28
n	7761.36	2148.50	15702.56	409.00	854.40	17499.30	102869.00
nbar	60.41	25.37	110.50	1.74	10.19	86.35	588.04
ncv	0.41	0.34	0.21	0.01	0.21	0.60	1.35
p.tx	0.55	0.50	0.10	0.40	0.49	0.74	0.80

The 10th and 90th percentile number of individuals, clusters and blocks are around: 850 to 17,500 (individuals), 30 to 495 (clusters), and 5 to 85 (blocks).

unit	10%	90%
n	854.4	17499.3
J	30.0	495.0
K	5.0	85.3

2.1 Which studies have singletons and doubletons?

This table lists the number of singletons and doubletons for the Blocked experiments

UniqueID	num_singletons	num_doubletons
Beacon_S8TGPA	27	13
Beacon_S8TOTCR	27	13
Beacon_SRPRST2	27	13

UniqueID	num_singletons	num_doubletons
EMERGE_team2__BB	2	0
EMERGE_eclsb2_BB	2	0
EMERGE_team2__BBSSR	2	0
EMERGE_eclsb2_BBSSR	2	0
CARES_DC_PT_Incred	36	8
CARES_TR_BPI_Incred	36	8
CARES_TR_WRS_Incred	36	8
CARES_DC_EMIT_PTH	36	8
CARES_DC_CST_COMPTOT_PTH	36	8
CARES_TR_SSRS_PTH	36	8
CARES_DC_PT_Tools_Mind	36	8
CARES_TR_WRS_Tools_Mind	36	8
CLC_GRCCSS_POST	14	1
CLC_PCCRERCORE	14	1
MathPD_MTSS	0	1
MathPD_MTSS_FRACTDEC	0	1
MathPD_MTSS_PCTRATIO	0	1
MPC_eclsb2_scale	1	25
MPC_wj_ap2	1	25
PowerTeaching_TSTSSTAND	0	2
Success_S14WJLWI_SCORE	0	4
TennStar_gktmathss	37	87
TennStar_gktreadss	37	87
TRIAD_rema2	2	3
TRIADScale-up_rema2	16	0
jg_crt_2	0	0
jg_crt_6	769	7
jg_crt_7	0	12
jg_crt_8	0	12
jg_crt_10	0	0
jg_crt_11	0	12
jg_crt_12	0	12
jg_crt_16	0	0
jg_crt_17	12	9
jg_crt_18	12	9
jg_crt_19	12	9
jg_crt_23	25	5
jg_crt_29	0	0
jg_crt_30	0	0
jg_crt_31	79	13
jg_crt_34	0	0
jg_crt_41	0	0
jg_crt_52	508	0
jg_crt_56	5	5
jg_crt_63	161	9
jg_crt_68	79	13
jg_crt_69	79	13

We can tally the outcomes by study type. Many of the studies had at least one singleton:

blocked	Collection	n_outcome	n_no_singletons	n_no_doubletons
Blocked	Curated	28	5	5
Blocked	Public	22	11	8
Unblocked	Public	38	38	38

2.2 Studies with low degrees of freedom

In this report, we define "Low" to be strictly less than 10 degrees of freedom, with degrees of freedom defined as

$$df_{cons} = J - 2K - 1$$

Here are all studies with 30 or fewer "conservative" degrees of freedom (so low and near low):

UniqueID	cons_df	n	J	K	Jbar	Jcv	ncov.1	ncov.2
TRIADScale-up_rema2	0	409	16	8	2.00	0.00	1	1
jg_crt_52	0	102869	508	254	2.00	0.00	2	2
CARES_DC_PT_Incred	8	1298	52	22	2.36	0.33	1	1
CARES_TR_BPI_Incred	8	1318	52	22	2.36	0.33	1	1
CARES_TR_WRS_Incred	8	1320	52	22	2.36	0.33	1	1
CARES_DC_EMIT_PTH	8	1229	52	22	2.36	0.33	1	1
CARES_DC_CST_COMPTOT_PTH	8	1206	52	22	2.36	0.33	1	1
CARES_TR_SSRS_PTH	8	1276	52	22	2.36	0.33	1	1
CARES_DC_PT_Tools_Mind	8	1268	52	22	2.36	0.33	1	1
CARES_TR_WRS_Tools_Mind	8	1293	52	22	2.36	0.33	1	1
jg_crt_23	9	926	41	16	2.56	0.43	2	2
jg_crt_63	9	8987	179	85	2.11	0.15	4	4
CLC_GRCCSS_POST	10	9557	28	9	3.11	0.78	2	2
CLC_PCCRERCORE	10	16465	28	9	3.11	0.78	2	2
jg_crt_7	16	1386	30	7	4.29	0.18	11	11
jg_crt_8	16	1335	30	7	4.29	0.18	11	11
jg_crt_11	16	2588	30	7	4.29	0.18	12	12
jg_crt_12	16	2612	30	7	4.29	0.18	12	12
TRIAD_rema2	24	1208	38	7	5.43	0.10	1	1
jg_crt_6	24	15993	802	389	2.06	0.42	15	15
MathPD_MTSS	27	2132	39	6	6.50	0.19	1	1
MathPD_MTSS_FRACTDEC	27	2132	39	6	6.50	0.19	1	1
MathPD_MTSS_PCTRATIO	27	2132	39	6	6.50	0.19	1	1
Success_S14WJLWI_SCORE	27	1631	37	5	7.40	0.74	1	1

3 Some statistics on our estimators

For each outcome we calculate a bunch of ATE estimates and SE estimates. We then look at the range of ATE estimates and the ratio of the max to min SE estimate.

We can then tally the number of outcomes with a large ranges of SE (2x or more) and/or ATE (0.10 or more):

big_ate	big_se	n
No	No	40
No	Yes	1
Yes	No	8
Yes	Yes	1

3.1 How often the ATE range is large

Here we count the number of ranges greater than 0.10 and 0.05:

group	measure	n	n_10	big_10	n_05	big_5
all	ate	50	9	0.18	19	0.38
estimand	ate_Cluster	50	3	0.06	11	0.22
estimand	ate_Person	50	3	0.06	8	0.16
noFI	ate_noFI	50	8	0.16	16	0.32
noFI_estimand	ate_Cluster_noFI	50	3	0.06	9	0.18
noFI_estimand	ate_Person_noFI	50	3	0.06	3	0.06

3.2 How often estimated Standard error ratio is large

The `n_1.2`, etc are the number of outcomes with a ratio of that amount or larger. The "big" is just the percent of outcomes.

group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2	max
all	se	50	29	0.58	8	0.16	2	0.04	3.46
estimand	se_Cluster	50	20	0.40	3	0.06	1	0.02	3.45
estimand	se_Person	50	17	0.34	5	0.10	0	0.00	1.75
noFI	se_noFI	50	20	0.40	5	0.10	0	0.00	1.71
noFI_estimand	se_Cluster_noFI	50	12	0.24	2	0.04	0	0.00	1.62
noFI_estimand	se_Person_noFI	50	12	0.24	3	0.06	0	0.00	1.56

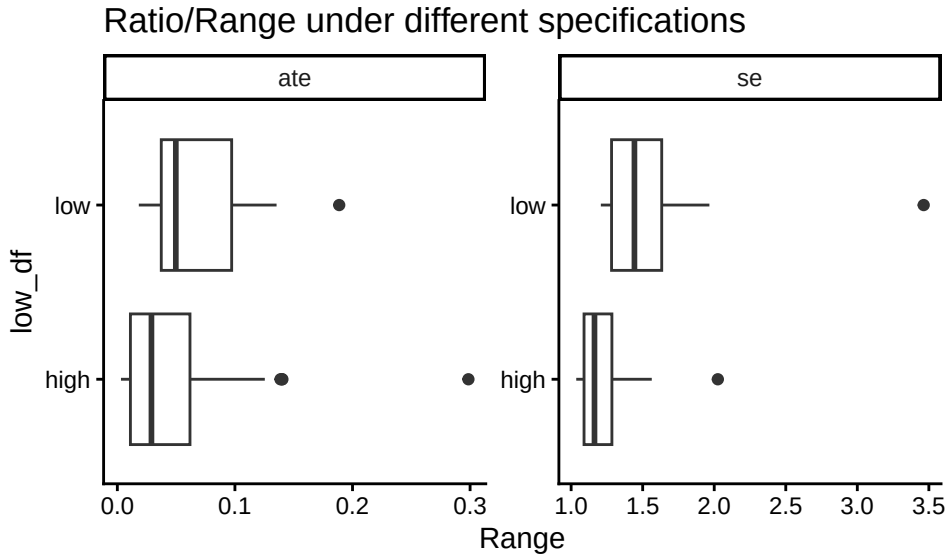
3.3 Stability connected to df

Here we divide our outcomes by low vs high degree of freedom, and then count how many ratios are large vs. small within those subgroups:

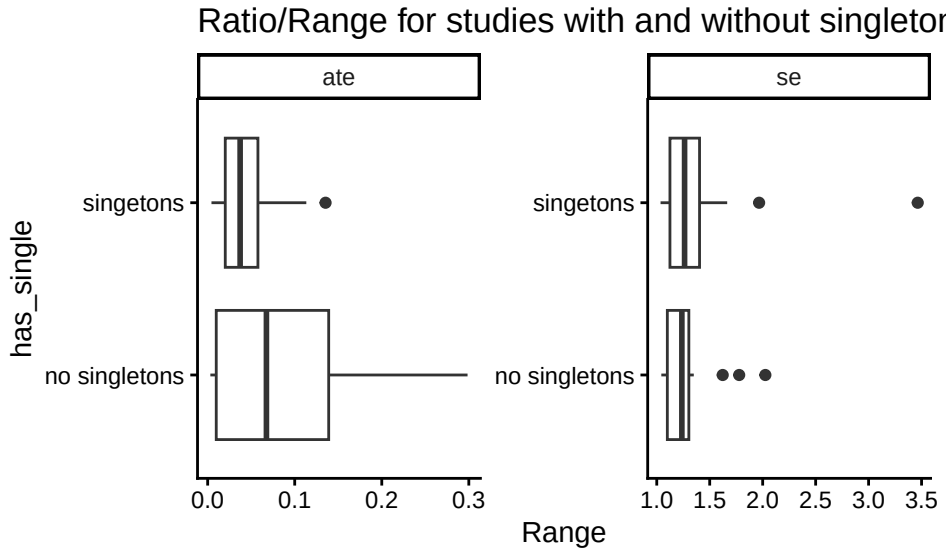
low_df	group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2
low	all	se	16	16	1.00	6	0.38	1	0.06
high	all	se	34	13	0.38	2	0.06	1	0.03
low	estimand	se_Cluster	16	13	0.81	2	0.12	1	0.06
high	estimand	se_Cluster	34	7	0.21	1	0.03	0	0.00
low	estimand	se_Person	16	6	0.38	4	0.25	0	0.00
high	estimand	se_Person	34	11	0.32	1	0.03	0	0.00

low_df	group	measure	n	n_1.2	big_1.2	n_1.5	big_1.5	n_2	big_2
low	noFI	se_noFI	16	13	0.81	3	0.19	0	0.00
high	noFI	se_noFI	34	7	0.21	2	0.06	0	0.00
low	noFI_estimand	se_Cluster_noFI	16	7	0.44	1	0.06	0	0.00
high	noFI_estimand	se_Cluster_noFI	34	5	0.15	1	0.03	0	0.00
low	noFI_estimand	se_Person_noFI	16	6	0.38	3	0.19	0	0.00
high	noFI_estimand	se_Person_noFI	34	6	0.18	0	0.00	0	0.00

We can also make a quick boxplot to show the difference in ratio by low vs high df:



And singleton vs not:



3.4 When are SE estimators defined?

In the paper we write about how singletons make SEs undefined, and how we can run out of degrees of freedom if we adjust for too many covariates. To support this, the following table counts the number of undefined estimates for each estimator, for different sets of baselind covariates included.

Estimator	n	n_ATE_all	n_ATE_best	n_ATE_none	n_SE_all	n_SE_best	n_SE_none
LRa-FE-db	50	0	0	0	2	0	0
LRa-FE-het	50	0	0	0	8	0	0
LRa-FIcw-db	50	12	2	0	40	34	34
LRa-FIcw-het	50	12	2	0	41	34	34
LRapw-FE-db	50	0	0	0	2	0	0
LRapw-FE-het	50	0	0	0	4	0	0
LRapw-FIpw-db	50	12	2	0	40	34	34
LRapw-FIpw-het	50	12	2	0	40	34	34
LRi-FE-crve	50	0	0	0	0	0	0
LRi-FE-db	50	0	0	0	2	0	0
LRi-FIpw-crve	50	10	0	0	40	34	34
LRi-FIpw-db	50	12	2	0	40	34	34
LRicw-FE-crve	50	0	0	0	0	0	0
LRicw-FE-db	50	0	0	0	2	0	0
LRicw-FIcw-crve	50	10	0	0	40	34	34
LRicw-FIcw-db	50	12	2	0	40	34	34
MLM-FE	50	0	0	0	0	0	0
MLM-FIcw	50	0	0	0	0	0	0
MLM-RE	50	0	0	0	0	0	0

The next table counts the number of estimates for each **outcome** that are undefined (the different columns correspond to adjusting for all covariates, a single best covariate, or no covariates):

UniqueID	n	n_ATE_all	n_ATE_best	n_SE_all	n_SE_best	n_SE_none
Beacon_S8TGPA	19	0	0	8	8	8
Beacon_S8TOTCR	19	0	0	8	8	8
Beacon_SRPRST2	19	0	0	8	8	8
CARES_DC_CST_COMPTOT_PTH	19	0	0	8	8	8
CARES_DC_EMIT_PTH	19	0	0	8	8	8
CARES_DC_PT_Incred	19	0	0	8	8	8
CARES_DC_PT_Tools_Mind	19	0	0	8	8	8
CARES_TR_BPI_Incred	19	0	0	8	8	8
CARES_TR_SSRS_PTH	19	0	0	8	8	8
CARES_TR_WRS_Incred	19	0	0	8	8	8
CARES_TR_WRS_Tools_Mind	19	0	0	8	8	8
CLC_GRCCSS_POST	19	8	0	13	8	8
CLC_PCCRERCORE	19	8	0	12	8	8
EMERGE_eclsb2_BB	19	0	0	8	8	8
EMERGE_eclsb2_BBSSR	19	0	0	8	8	8
EMERGE_team2__BB	19	0	0	8	8	8
EMERGE_team2__BBSSR	19	0	0	8	8	8
MPC_eclsb2_scale	19	0	0	8	8	8
MPC_wj_ap2	19	0	0	8	8	8
MathPD_MTSS	19	8	0	10	0	0
MathPD_MTSS_FRACTDEC	19	8	0	10	0	0
MathPD_MTSS_PCTRATIO	19	8	0	9	0	0
PowerTeaching_TSTSSTAND	19	8	0	8	0	0
Success_S14WJLWI_SCORE	19	0	0	0	0	0
TRIADScale-up_rema2	19	6	6	8	8	8
TRIAD_rema2	19	0	0	8	8	8
TennStar_gktmathss	19	8	0	9	8	8

UniqueID	n	n_ATE_all	n_ATE_best	n_SE_all	n_SE_best	n_SE_none
TennStar_gktreadss	19	8	0	9	8	8
jg_crt_10	19	0	0	0	0	0
jg_crt_11	19	0	0	0	0	0
jg_crt_12	19	0	0	0	0	0
jg_crt_16	19	0	0	0	0	0
jg_crt_17	19	0	0	10	8	8
jg_crt_18	19	0	0	9	8	8
jg_crt_19	19	0	0	8	8	8
jg_crt_2	19	0	0	0	0	0
jg_crt_23	19	0	0	8	8	8
jg_crt_29	19	0	0	2	0	0
jg_crt_30	19	0	0	0	0	0
jg_crt_31	19	0	0	8	8	8
jg_crt_34	19	0	0	0	0	0
jg_crt_41	19	0	0	0	0	0
jg_crt_52	19	6	6	8	8	8
jg_crt_56	19	0	0	8	8	8
jg_crt_6	19	0	0	8	8	8
jg_crt_63	19	0	0	8	8	8
jg_crt_68	19	0	0	8	8	8
jg_crt_69	19	0	0	8	8	8
jg_crt_7	19	8	0	8	0	0
jg_crt_8	19	8	0	8	0	0

Note for no covariate adjustment, we see that 8 estimators will not give SEs for the studies that have singletons (we have 8 of those estimators).

So these two tables count undefined-ness first by estimator and then by outcome.

4 ATE Ranges and SE ratios by study

We can look at specific outcomes to see the range of ATE estimates and the ratio of SE estimates for that outcome. The "N" column shows number of defined estimates comprising the given range.

UniqueID	num_singletons	Range	ATE_N	Ratio	SE_Min	SE_Max	SE_N
jg_crt_2	0	0.299	19	2.025	0.061	0.125	19
jg_crt_8	0	0.189	19	1.622	0.067	0.108	19
MathPD_MTSS	0	0.140	19	1.228	0.122	0.149	19
MathPD_MTSS_PCTRATIO	0	0.140	19	1.190	0.116	0.138	19
jg_crt_11	0	0.139	19	1.286	0.084	0.108	19
CARES_TR_BPI_Incred	36	0.135	19	1.263	0.124	0.157	11
MathPD_MTSS_FRACTDEC	0	0.125	19	1.246	0.118	0.147	19
jg_crt_7	0	0.122	19	1.778	0.046	0.081	19
CARES_TR_WRS_Incred	36	0.113	19	1.262	0.109	0.138	11
CARES_TR_SSRS_PTH	36	0.092	19	1.665	0.082	0.136	11
PowerTeaching_TSTSSTAND	0	0.088	19	1.287	0.048	0.061	19
jg_crt_56	5	0.088	19	1.177	0.095	0.112	11
CARES_DC_CST_COMPTOT_PTH	36	0.081	19	1.292	0.064	0.082	11
EMERGE_eclsb2_BB	2	0.077	19	1.058	0.101	0.107	11
EMERGE_team2___BB	2	0.062	19	1.115	0.121	0.135	11
CLC_GRCCSS_POST	14	0.062	19	1.582	0.038	0.060	11

UniqueID	num_singletons	Range	ATE_N	Ratio	SE_Min	SE_Max	SE_N
TRIAD_rema2	2	0.060	19	1.397	0.075	0.105	11
CARES_TR_WRS_Tools_Mind	36	0.052	19	1.212	0.089	0.108	11
MPC_wj_ap2	1	0.051	19	1.174	0.070	0.082	11
CLC_PCCRERCORE	14	0.048	19	1.357	0.036	0.048	11
jg_crt_12	0	0.047	19	1.350	0.080	0.108	19
EMERGE_eclsb2_BBSSR	2	0.044	19	1.095	0.118	0.129	11
MPC_eclsb2_scale	1	0.041	19	1.122	0.084	0.094	11
TRIADScale-up_rema2	16	0.041	13	1.450	0.106	0.154	11
CARES_DC_PT_Incred	36	0.039	19	1.207	0.073	0.088	11
jg_crt_16	0	0.039	19	1.253	0.067	0.084	19
CARES_DC_PT_Tools_Mind	36	0.038	19	1.435	0.063	0.091	11
EMERGE_team2__BBSSR	2	0.037	19	1.200	0.114	0.136	11
CARES_DC_EMIT_PTH	36	0.036	19	1.465	0.051	0.075	11
jg_crt_19	12	0.036	19	1.563	0.027	0.042	11
jg_crt_69	79	0.022	19	1.076	0.031	0.034	11
TennStar_gktmathss	37	0.022	19	1.083	0.045	0.048	11
jg_crt_52	508	0.021	13	3.464	0.026	0.089	11
jg_crt_6	769	0.021	19	1.348	0.015	0.020	11
jg_crt_68	79	0.021	19	1.283	0.026	0.034	11
jg_crt_17	12	0.020	19	1.319	0.022	0.030	11
jg_crt_23	25	0.019	19	1.966	0.051	0.099	11
jg_crt_63	161	0.018	19	1.289	0.027	0.035	11
jg_crt_34	0	0.018	19	1.043	0.074	0.077	19
TennStar_gktreadss	37	0.015	19	1.130	0.042	0.048	11
jg_crt_18	12	0.015	19	1.407	0.021	0.030	11
jg_crt_31	79	0.010	19	1.135	0.022	0.025	11
jg_crt_41	0	0.010	19	1.057	0.074	0.078	19
Success_S14WJLWI_SCORE	0	0.010	19	1.152	0.055	0.064	19
Beacon_S8TOTCR	27	0.009	19	1.060	0.039	0.041	11
Beacon_S8TGPA	27	0.006	19	1.036	0.046	0.047	11
jg_crt_10	0	0.005	19	1.090	0.047	0.051	19
jg_crt_29	0	0.005	19	1.091	0.014	0.015	19
Beacon_SRPRST2	27	0.004	19	1.043	0.044	0.046	11
jg_crt_30	0	0.003	19	1.103	0.015	0.016	19

The study with the median ratio:

UniqueID	num_singletons	Range	ATE_N	Ratio	SE_Min	SE_Max	SE_N
MathPD_MTSS_FRACTDEC	0	0.13	19	1.25	0.12	0.15	19

4.1 CARES_DC_CST_COMPTOT_PTH study details

We use this study as a given example in the text:

Estimator	Estimand	ATE_hat_std	SE_hat_std
LRi-FE-db	Person	0.151	0.064
MLM-RE	Cluster	0.157	0.064
LRapw-FE-db	Person	0.146	0.065
LRi-FE-crve	Person	0.151	0.065

Estimator	Estimand	ATE_hat_std	SE_hat_std
MLM-FE	Cluster	0.148	0.065
LRapw-FE-het	Person	0.146	0.067
LRa-FE-db	Cluster	0.076	0.073
MLM-FIcw	Cluster	0.076	0.077
LRa-FE-het	Cluster	0.076	0.077
LRicw-FE-db	Cluster	0.080	0.080
LRicw-FE-crve	Cluster	0.080	0.082
LRa-FIcw-db	Cluster	0.080	NA
LRa-FIcw-het	Cluster	0.080	NA
LRapw-FIpw-db	Person	0.149	NA
LRapw-FIpw-het	Person	0.149	NA
LRi-FIpw-crve	Person	0.146	NA
LRi-FIpw-db	Person	0.146	NA
LRicw-FIcw-crve	Cluster	0.080	NA
LRicw-FIcw-db	Cluster	0.080	NA

The ratio for this study:

UniqueID	num_singletons	Range	ATE_N	Ratio	SE_Min	SE_Max	SE_N
CARES_DC_CST_COMPTOT_PTH	36	0.08	19	1.29	0.06	0.08	11

4.2 What is the average range and ratio?

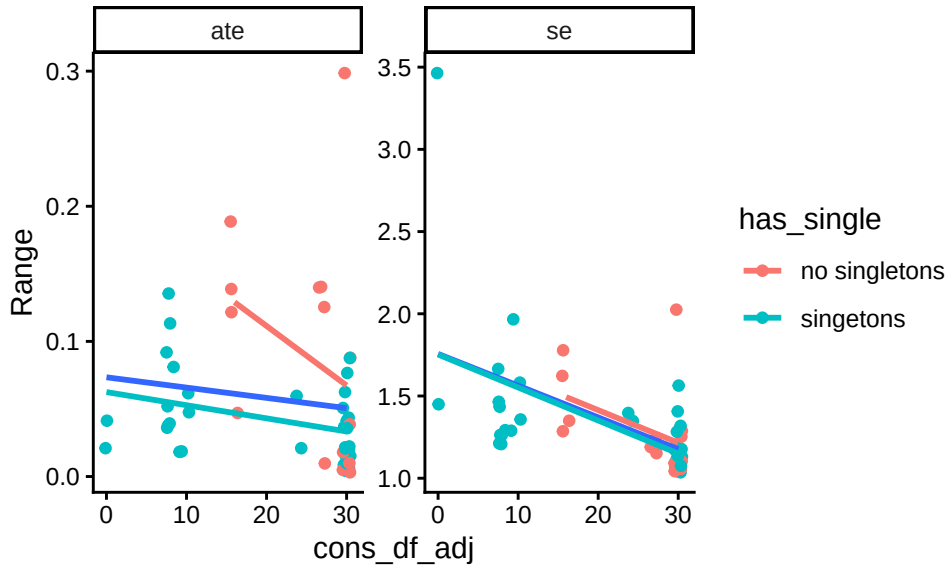
The following table shows the average range (for ATEs) and average ratio (for estimated SEs) across all our outcomes:

This tells us, for example, that "On average, the largest standard error is around 33% larger than the smallest, across studies." This is for the "best" covariate adjustment:

group	Estimand	measure	avg	sd	Q05	Q95	n
all	Overall	ate	0.06	0.06	0.00	0.14	50
estimand	Cluster	ate_Cluster	0.03	0.04	0.00	0.10	50
estimand	Person	ate_Person	0.03	0.04	0.00	0.11	50
all	Overall	se	1.33	0.38	1.05	1.88	50
estimand	Cluster	se_Cluster	1.25	0.36	1.04	1.56	50
estimand	Person	se_Person	1.17	0.18	1.01	1.55	50

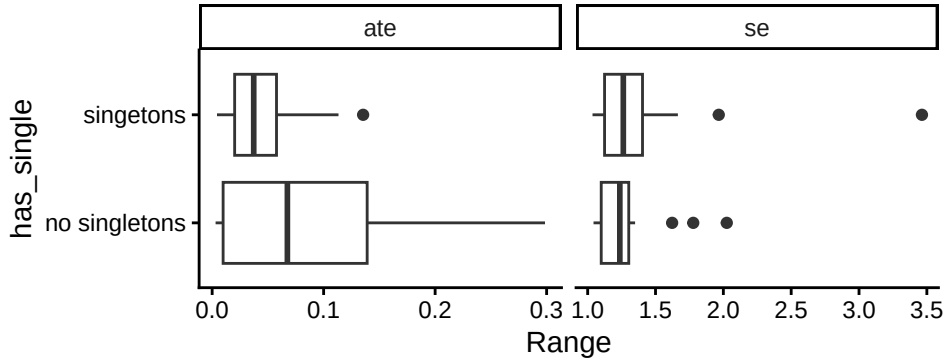
4.3 ATE range and SE ratios vs study characteristics

We can see if our "conservative df" for each study predicts the range of point estimates or ratio of SE estimates. We top-code all studies with 30 df or more to 30 to make the plots easier to read.



Commentary: This plot does show there is some tendency for ranges and ratios to go down as DF goes up, but there is a lot of variation around that tendency.

We next look at the distribution of range/ratio of estimates across outcome, grouping by singleton status. We see the studies with singletons tend to have smaller ATE ranges, and the range of SE estimates is marginally higher—note that the singleton studies SE ratios are over fewer estimators (none of the FI estimators are defined), which would generally push the ratios *lower* as there are fewer points to take the max and min from.



Commentary: I am not sure if these plots are useful or informative. I am not sure what they mean, either.

5 Variation in the ATE Estimates

5.1 Studies with a high range (more than 0.15) of ATE estimates

Project	UniqueID	K	J	n	cons_df	R_ate	R_ate_Clu	R_ate_Per
Kim/set_2	jg_crt_2	7	83	2174	69	0.30	0.20	0.05
Kim/set_7	jg_crt_8	7	30	1335	16	0.19	0.11	0.18

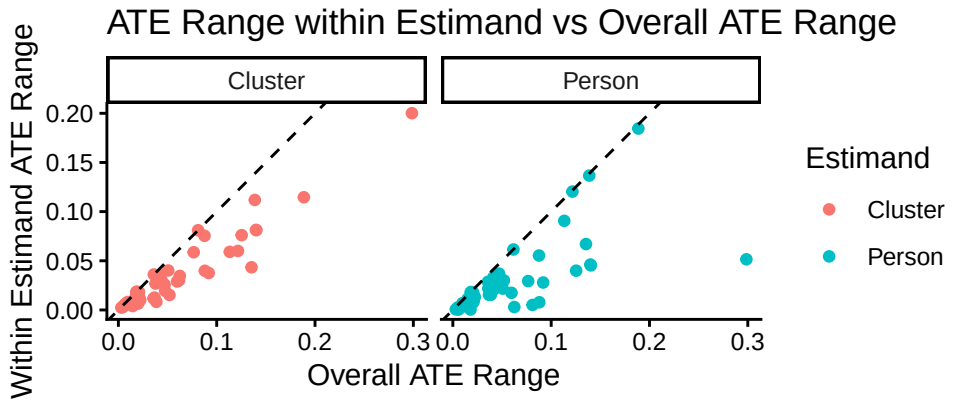
5.2 Range within Estimand

If we go within study, we get the following average reduction in the ATE range when we go to within estimand from overall:

Estimand	n	mean_reduction	sd_reduction	cor	mean_del
Cluster	50	0.64	0.21	-0.14	0.02
Person	50	0.52	0.28	-0.01	0.03

We see a notable drop in the ranges when we go to "within estimand" from "overall."

We can plot the ranges within vs. between:

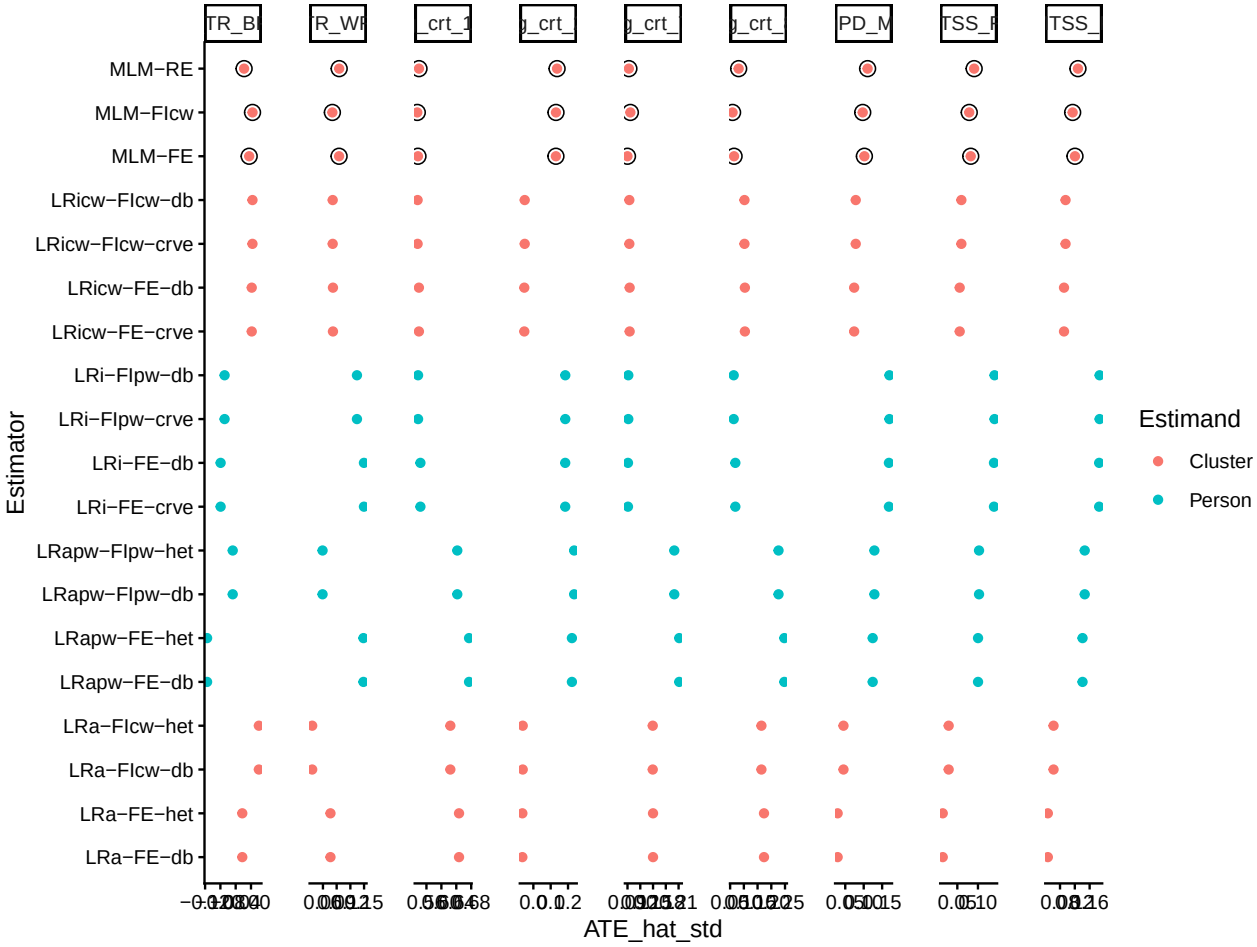


If we look at all studies with an ATE range above 0.10 for any estimand, we get the following list:

Project	UniqueID	K	J	n	cons_df	R_ate	R_ate_Clu	R_ate_Per
CARES	CARES_TR_BPI_Incred	22	52	1318	8	0.14	0.04	0.07
CARES	CARES_TR_WRS_Incred	22	52	1320	8	0.11	0.06	0.09
Math PD	MathPD_MTSS	6	39	2132	27	0.14	0.08	0.05
Math PD	MathPD_MTSS_FRACTDEC	6	39	2132	27	0.13	0.08	0.04
Math PD	MathPD_MTSS_PCTRATIO	6	39	2132	27	0.14	0.08	0.05
Kim/set_2	jg_crt_2	7	83	2174	69	0.30	0.20	0.05
Kim/set_7	jg_crt_7	7	30	1386	16	0.12	0.06	0.12
Kim/set_7	jg_crt_8	7	30	1335	16	0.19	0.11	0.18
Kim/set_9	jg_crt_11	7	30	2588	16	0.14	0.11	0.14

Some statistics on the selected studies:

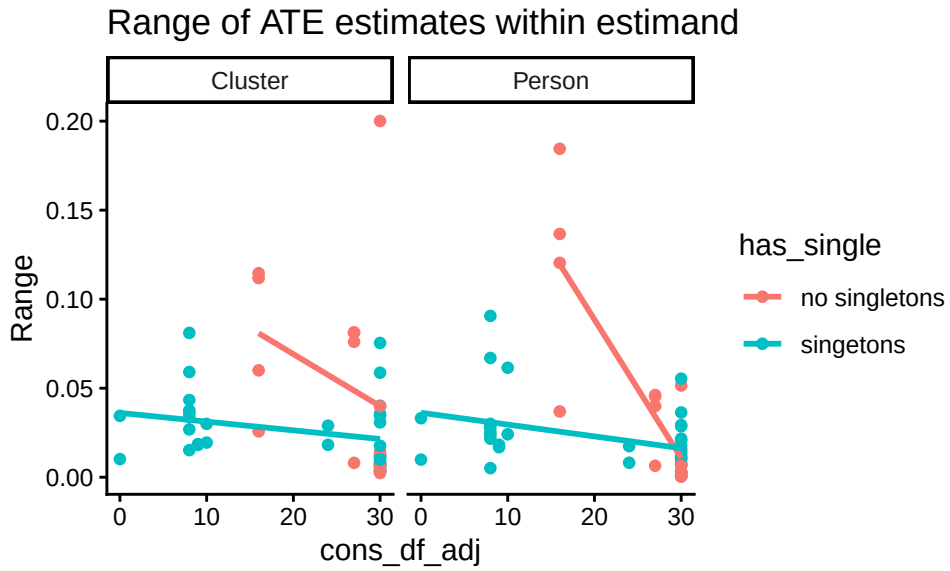
UniqueID	num_singletons	ncv	Jcv	ATE_min	ATE_max	range	SE_avg	n_est	n_def
CARES_TR_BPI_Incred	36	0.53	0.33	-0.11	0.02	0.14	0.14	19	19
CARES_TR_WRS_Incred	36	0.52	0.33	0.04	0.15	0.11	0.12	19	19
MathPD_MTSS	0	0.44	0.19	0.03	0.17	0.14	0.13	19	19
MathPD_MTSS_FRACTDEC	0	0.44	0.19	0.01	0.14	0.13	0.13	19	19
MathPD_MTSS_PCTRATIO	0	0.44	0.19	0.05	0.19	0.14	0.12	19	19
jg_crt_11	0	0.35	0.18	0.54	0.67	0.14	0.10	19	19
jg_crt_2	0	1.35	0.41	-0.06	0.24	0.30	0.10	19	19
jg_crt_7	0	0.51	0.18	0.09	0.21	0.12	0.07	19	19
jg_crt_8	0	0.52	0.18	0.06	0.25	0.19	0.09	19	19



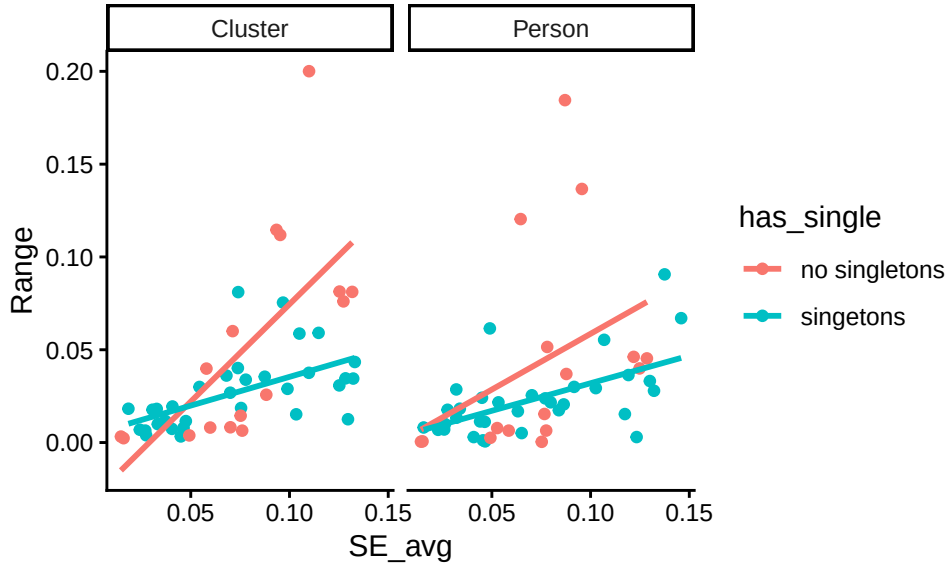
Note: The MLM estimators are circled to indicate they are not unbiased for either estimand, in principle.

5.3 What predicts range within estimand

Extending the overall plots from above, we can look at what study characteristics drive larger or smaller ranges, within estimand:

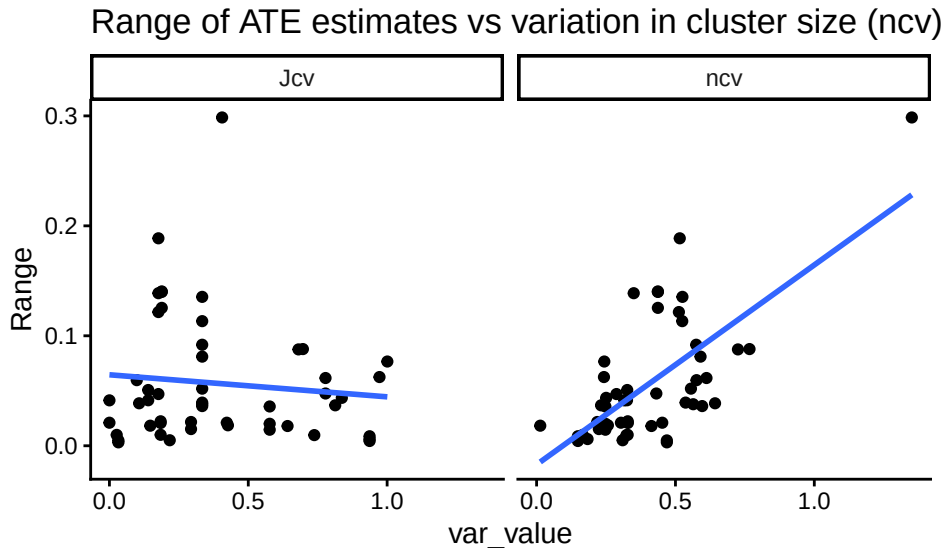


Range does go up as expected with average SE increasing:



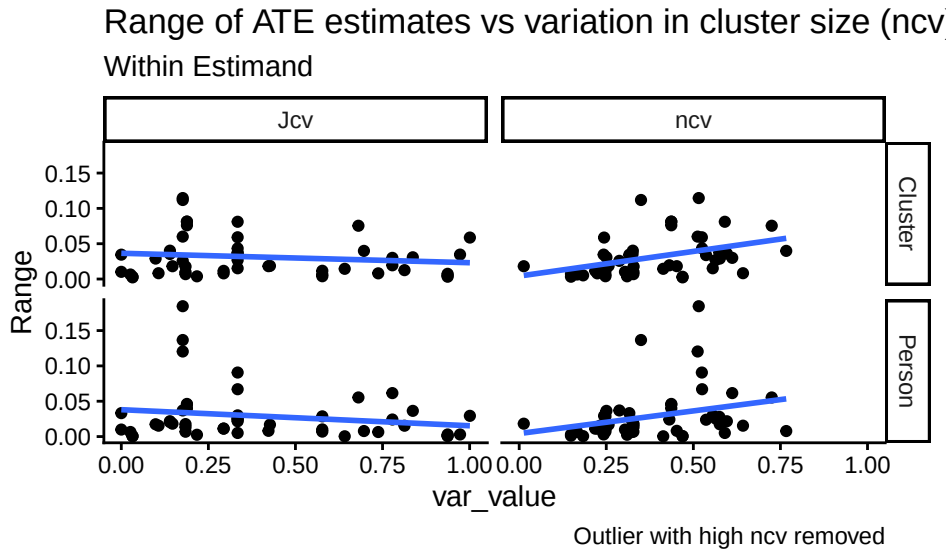
5.4 Does variation in cluster size or number of clusters predict range?

We can see if range depends on variation in block or cluster size:



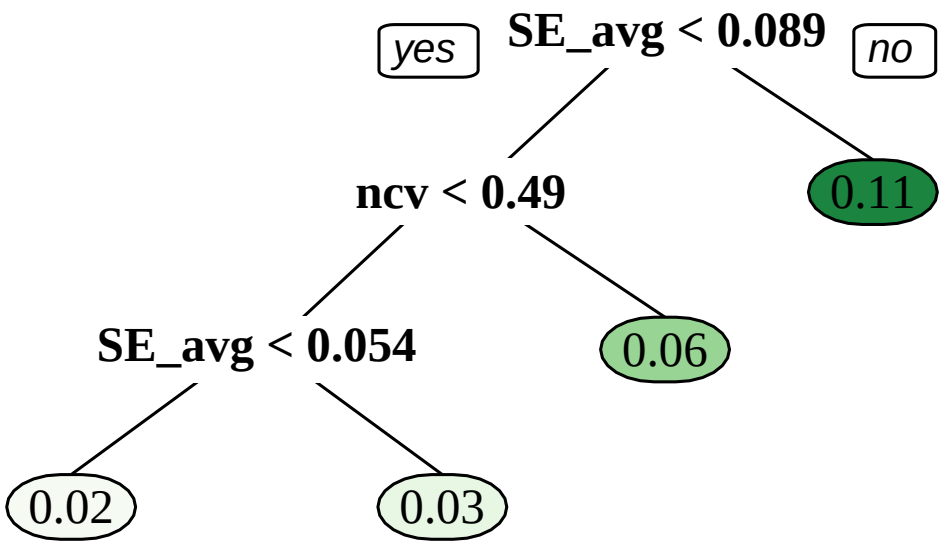
We do see the range go up as cluster size variation goes up, but not so much with number of clusters variation.

Within estimand, we can do the same as above (I dropped the extreme outlier above 1 for better visualization and more robust trend detection):

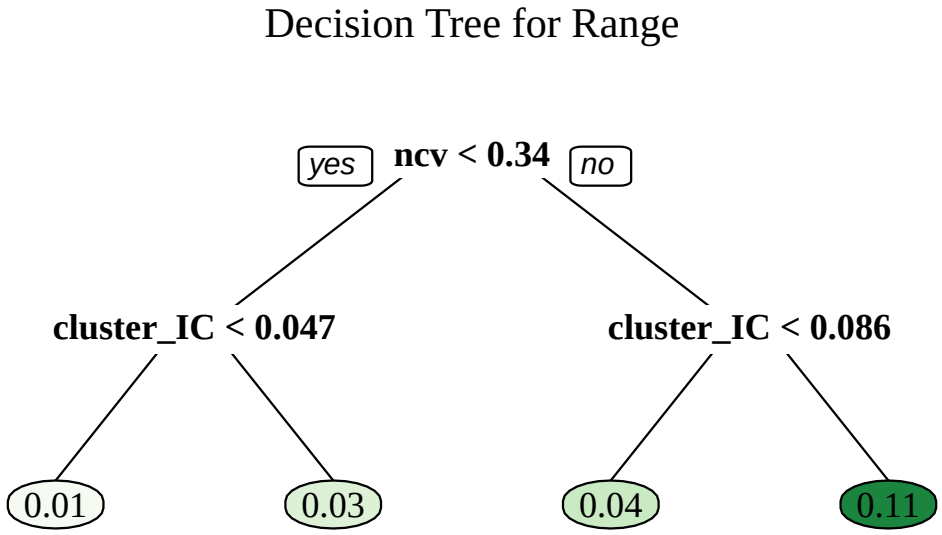


Within estimand, we see the same pattern with cluster size variation, but not number of clusters variation.

We can use trees to predict what predicts larger range:

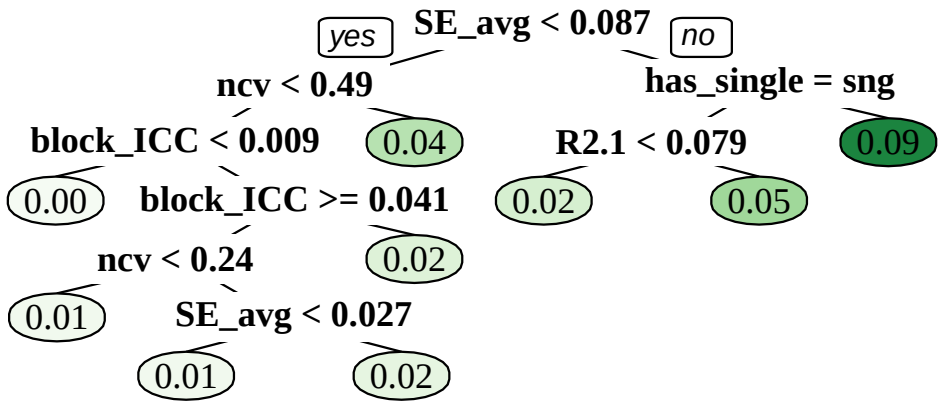


Big SE means big ranges. If we drop the average SE as a predictive feature, we get this:



These trees shows that uncertainty drives larger differences, and also that greater variation in cluster size drives larger differences, especially if there is variation in the cluster means.

If we look within estimand (i.e., fit our tree to the full set of 50 person ranges and 50 cluster ranges), we see that our datasets with larger R2.1 and no singletons have larger ranges (assuming the SEs are generally large):



5.5 Comparing low DF to high DF studies

We can look to see if studies with high or low df have generally larger ranges:

low_df	avg_rng	sd_rng	Q05	Q95	n	ratio
high	0.05	0.06	0.00	0.14	34	1.00
low	0.07	0.05	0.02	0.15	16	1.36

The low df studies tend to have higher ranges than the high df studies. Low df studies also are higher uncertainty studies, so this makes sense.

6 Variation in the Estimated Standard Errors

6.1 Studies with a high ratio of SE estimates

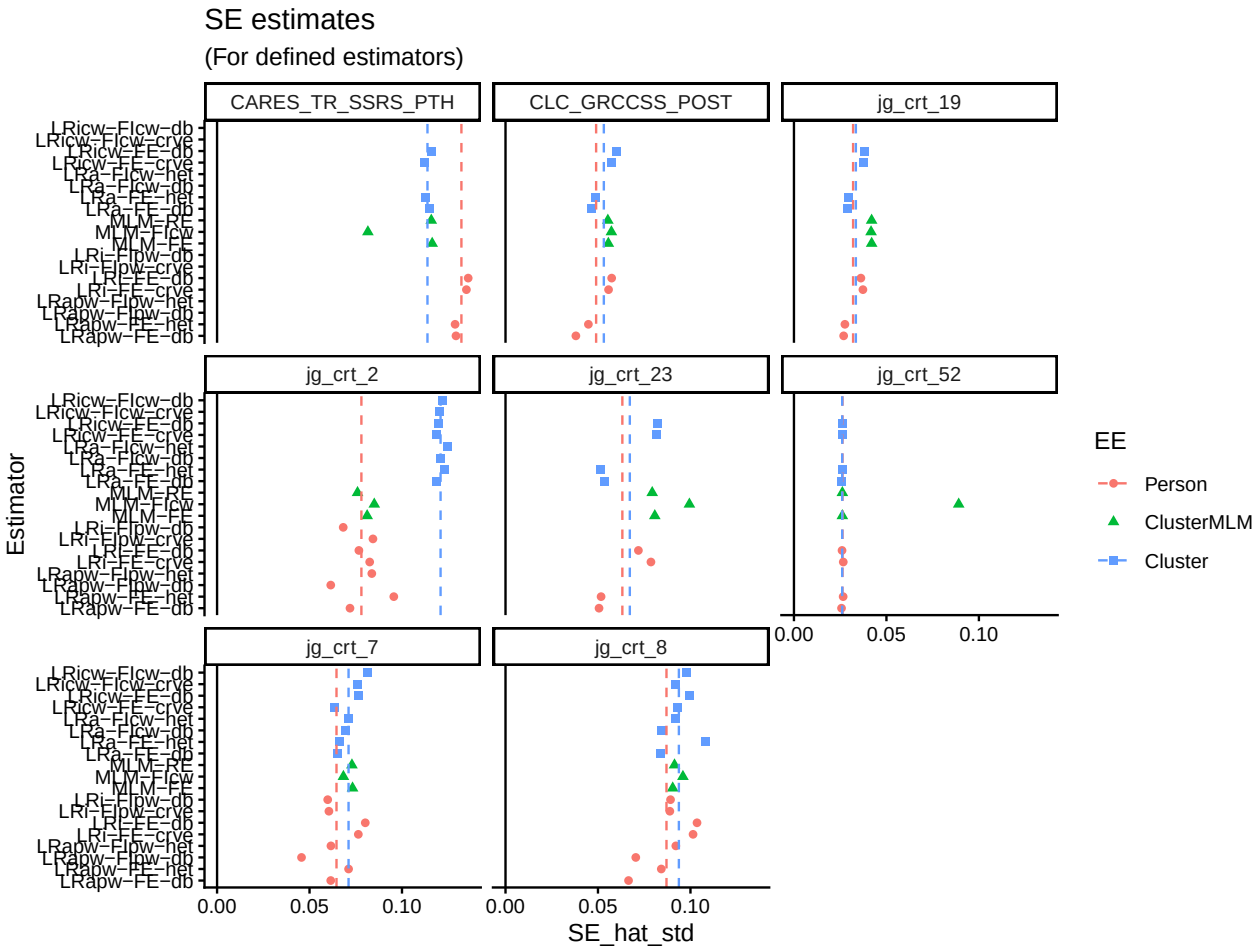
Here are all the studies with a min-max SE ratio above 1.5.

Project	UniqueID	K	J	n	cons_df	R_se	R_se_Clu	R_se_Per
CARES	CARES_TR_SSRS_PTH	11	52	1276	8	1.66	1.43	1.06
CLC	CLC_GRCCSS_POST	9	28	9557	10	1.58	1.30	1.51
Duflo/set_12	jg_crt_19	38	495	27056	419	1.56	1.45	1.39
Kim/set_2	jg_crt_2	7	83	2174	69	2.03	1.64	1.55
Bang/set_16	jg_crt_23	16	41	926	9	1.97	1.94	1.56
Abaluck/set_35	jg_crt_52	254	508	102869	0	3.46	3.45	1.04
Kim/set_7	jg_crt_7	7	30	1386	16	1.78	1.28	1.75
Kim/set_7	jg_crt_8	7	30	1335	16	1.62	1.29	1.56

Some extra info on these studies:

UniqueID	cons_df	num_singletons	SE_min	SE_max	n_est	n_def
CARES_TR_SSRS_PTH	8	36	0.0816	0.1358	19	11
CLC_GRCCSS_POST	10	14	0.0380	0.0602	19	11
jg_crt_19	419	12	0.0269	0.0420	19	11
jg_crt_2	69	0	0.0615	0.1245	19	19
jg_crt_23	9	25	0.0506	0.0994	19	11
jg_crt_52	0	508	0.0257	0.0891	19	11
jg_crt_7	16	0	0.0457	0.0812	19	19
jg_crt_8	16	0	0.0665	0.1079	19	19

The individual SE estimates for the troublesome studies:



6.2 How often are the SE estimates defined?

Here we look at 'best' and see how often the SE estimates are defined for each outcome. We see that for 34 outcomes, 57% of our estimators are defined and for 16 outcomes all of them are defined.

```
##
##        0.578947368421053  1
##   best                34 16
```

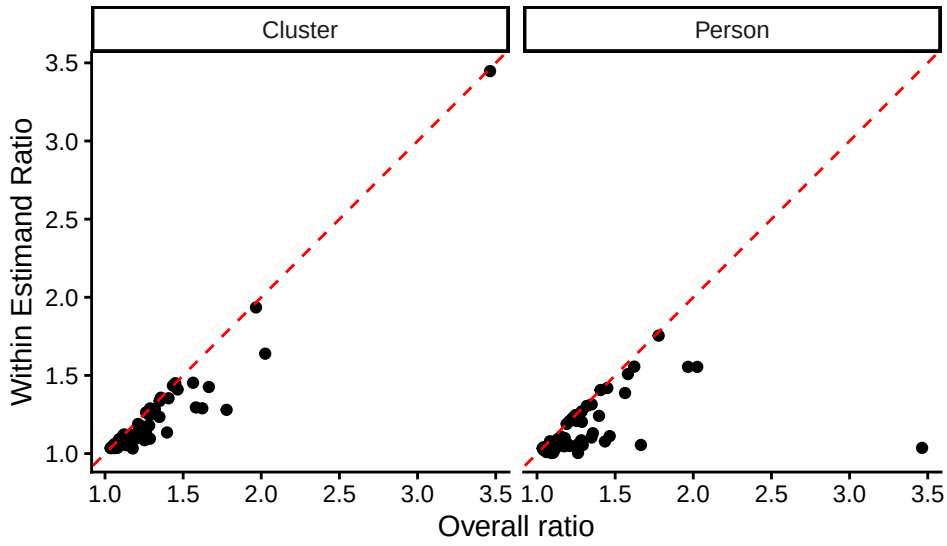
6.3 How much do SE ratios go down within Estimand?

In the following we look at how much the SE ratios shrink going within estimand.

Estimand	mean_reduction	sd_reduction
Cluster	0.94	0.07
Person	0.90	0.12

Overall, we are not seeing much of a reduction in the max/min SE ratio by going from overall to within.

We can look at the individual reductions:



7 Are SEs for cluster weight smaller than individual?

We might think the SEs for cluster-weighted estimands might be systematically larger than those for individual-weighted estimands. We check this on the empirical estimates.

To investigate, we fit a multilevel model to our raw estimated standard errors, having a random intercept for the outcome and for the estimator. We also allow for a "random slope" for estimand within outcome, as it is possible that some studies are better at targeting person vs cluster average effects. We finally allow a random slope by estimator for studies with singletons, to see if some estimators have estimated SEs that change more depending on whether there are singletons present. We further adjust for aspects of the study. Our main coefficient is the coefficient for Estimand:

```
## lmer(formula = log(SE_hat_std) ~ Estimand + Covariates + has_single +
##     I(log(1 + cons_df)) + I(log(nbar)) + I(log(J)) + (1 + has_single |
##     Estimator) + (1 + Estimand | UniqueID), data = combined)
##                       coef.est coef.se
## (Intercept)            0.24     0.27
## EstimandPerson        -0.03     0.03
## Covariatesnone         0.17     0.02
## Covariatesall         -0.10     0.02
## has_singlesingetons -0.22     0.10
## I(log(1 + cons_df))    0.00     0.04
## I(log(nbar))          -0.25     0.04
## I(log(J))             -0.44     0.05
##
## Error terms:
##  Groups    Name                 Std.Dev. Corr
##  UniqueID  (Intercept)          0.28
##            EstimandPerson       0.02     -1.00
##  Estimator (Intercept)          0.01
##            has_singlesingetons 0.05      1.00
##  Residual                       0.41
## ---
## number of obs: 1965, groups: UniqueID, 50; Estimator, 19
## AIC = 2302, DIC = 2193.3
## deviance = 2232.7
```

We see no evidence that targeting person estimands is different from cluster estimands in terms of precision. There is some variation in average uncertainty by estimator. The random slope standard deviation of 0.02 suggests some studies might have a difference between person and cluster.

We can look at the exponentiated coefficients to see the multiplicative effect on the SE:

	x
(Intercept)	1.27
EstimandPerson	0.97
Covariatesnone	1.18
Covariatesall	0.90
has_singlesingetons	0.80
I(log(1 + cons_df))	1.00
I(log(nbar))	0.78
I(log(J))	0.65

This suggests that person-weighted is 3% more precise (but the result is not statistically significant).

We can use a LR test to see if there is evidence of variation by study:

```
## Data: combined
## Models:
## Mnull: log(SE_hat_std) ~ Estimand + Covariates + has_single + I(log(1 + cons_df)) + I(log(nbar)) + I
## M: log(SE_hat_std) ~ Estimand + Covariates + has_single + I(log(1 + cons_df)) + I(log(nbar)) + I(log
##       npar    AIC    BIC  logLik -2*log(L) Chisq Df Pr(>Chisq)
## Mnull   13 2259.4 2332.0 -1116.7    2233.4
## M       15 2262.7 2346.4 -1116.3    2232.7 0.781  2     0.6767
```

No evidence of variation.

7.1 Analysis: What drives overall uncertainty?

Our model, above, tells us more about what drives overall uncertainty.

For example, relative to 'best', not controlling for any baseline covariates means estimated SEs are 17% larger. Controlling for everything makes the estimated SEs 10% smaller. Bigger studies in terms of either n or J have smaller SEs, as expected. The estimated SEs for the studies with singletons are generally smaller; it is unclear what that is coming from.

We can dig into which estimators are better or worse, in terms of their estimated SEs, by looking at the Empirical Bayes estimates of their random intercepts:

estimator	avg	singletons
LRa-FE-db	1.00	1.00
LRa-FE-het	1.00	1.01
LRa-FIcw-db	1.00	1.00
LRa-FIcw-het	1.00	1.00
LRapw-FE-db	1.00	0.98
LRapw-FE-het	1.00	1.01
LRapw-FIpw-db	1.00	0.99
LRapw-FIpw-het	1.00	1.00
LRi-FE-crve	1.00	0.99
LRi-FE-db	1.00	1.02
LRi-FIpw-crve	1.00	1.00
LRi-FIpw-db	1.00	1.00

estimator	avg	singletons
LRicw-FE-crve	0.98	0.89
LRicw-FE-db	1.01	1.04
LRicw-FIcw-crve	1.00	1.00
LRicw-FIcw-db	1.00	1.01
MLM-FE	1.00	1.01
MLM-FIcw	1.01	1.05
MLM-RE	1.00	1.00

The estimators have no variation in the average estimated SE, but do differ when faced with studies with singletons. In particular, MLM-FIcw is 6% larger, and LRicw-FE-crve is 12% smaller—the latter is likely due to poorly estimated CRVEs when there are singletons.

7.2 Which estimators tend to be most extreme?

Here we tag when each estimator is either the 1st or second smallest or 1st or second largest SE estimate of all availables for that outcome (within the "best" covariate set).

Estimator	Estimand	n_def	n_extreme	per
LRa-FIcw-db	Cluster	7	0	0.00
LRi-FIpw-crve	Person	7	0	0.00
MLM-FE	Cluster	25	2	0.08
MLM-RE	Cluster	25	4	0.16
LRa-FE-db	Cluster	25	5	0.20
LRi-FE-crve	Person	25	5	0.20
LRa-FIcw-het	Cluster	7	2	0.29
LRapw-FIpw-het	Person	7	2	0.29
LRi-FIpw-db	Person	7	2	0.29
LRicw-FIcw-crve	Cluster	7	2	0.29
LRicw-FIcw-db	Cluster	7	2	0.29
LRapw-FE-het	Person	25	8	0.32
LRa-FE-het	Cluster	25	9	0.36
LRicw-FE-crve	Cluster	25	9	0.36
LRicw-FE-db	Cluster	25	9	0.36
LRi-FE-db	Person	25	10	0.40
MLM-FIcw	Cluster	25	11	0.44
LRapw-FE-db	Person	25	14	0.56
LRapw-FIpw-db	Person	7	4	0.57

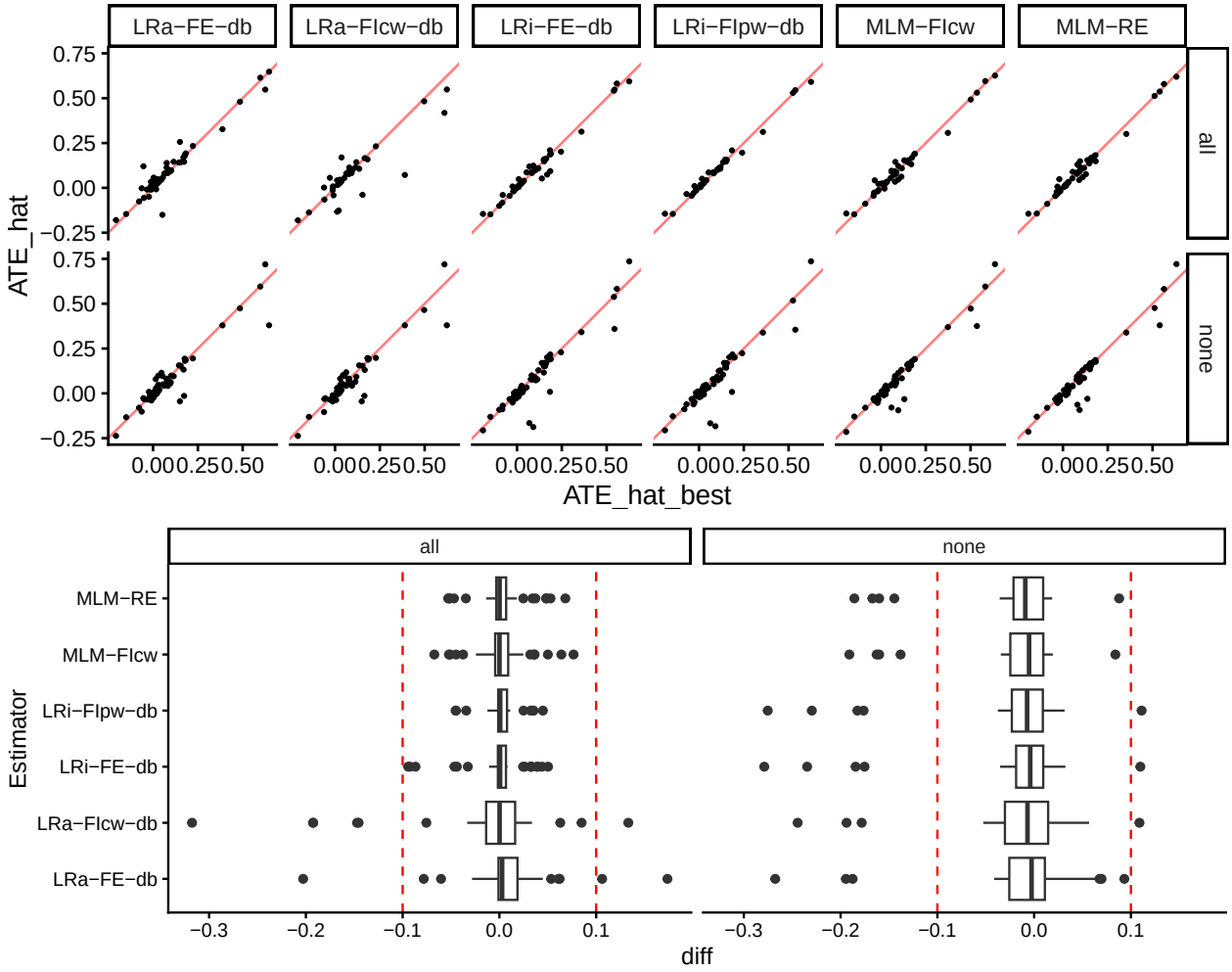
The `n_def` column is the number of outcomes where the estimator was even defined.

I'm not sure this table is particularly informative, but it does show that some estimators are more likely to be extreme than others. The problem is some estimators aren't even defined, which robs them of the opportunity to be informative.

8 How do covariates move the point estimates?

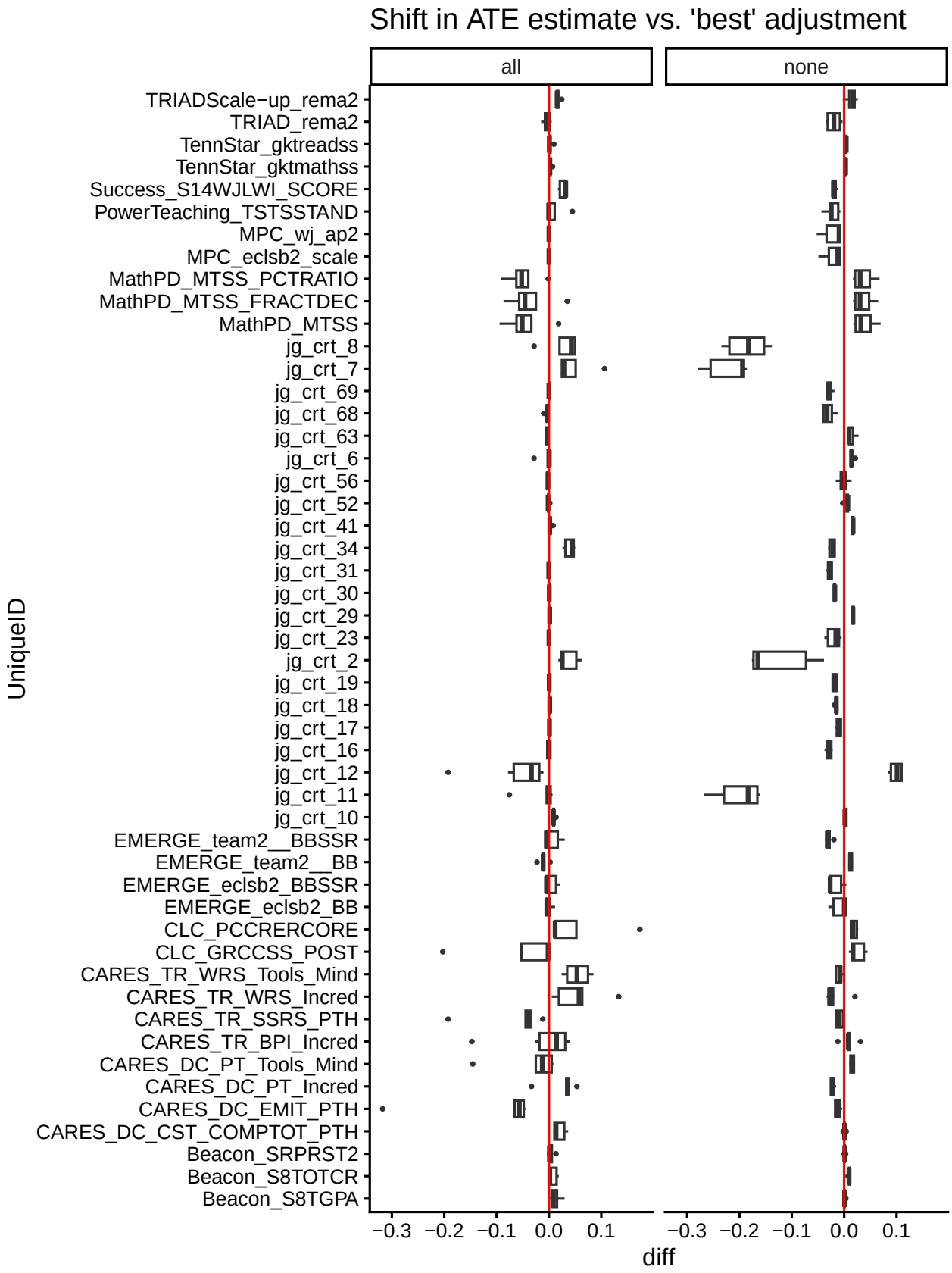
We next look at how much the point estimates change as we adjust for more covariates (all) or no covariates (none) vs. our baseline of "best". For each outcome and estimator we calculated the difference between the ATE estimate when controlling for the single best covariate, and the ATE estimate when controlling for nothing ("none") or all available covariates ("all"). We subset to the core estimators, removing redundancies and similar estimators

The first plot shows the scatterplot of the "best" ATE estimate vs. the "all" or "none" ATE estimate, faceted by estimator. The second plot shows the distribution of the difference between the "best" ATE estimate and the "all" or "none" ATE estimate, again grouped by estimator.



The plot above shows that, while generally adjusting for covariates does not change the estimate much, it can move a point estimate a very large amount. (The red lines are at 0.10 and -0.10.)

We can then look at shifts by study, rather than by estimator:



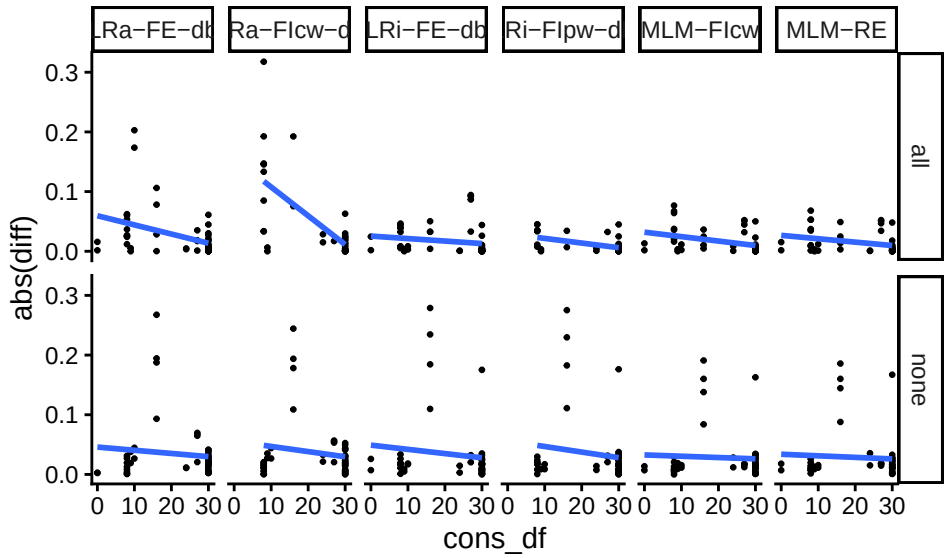
We see some studies tend to have large impacts of adjustment across the estimators, and other studies not so

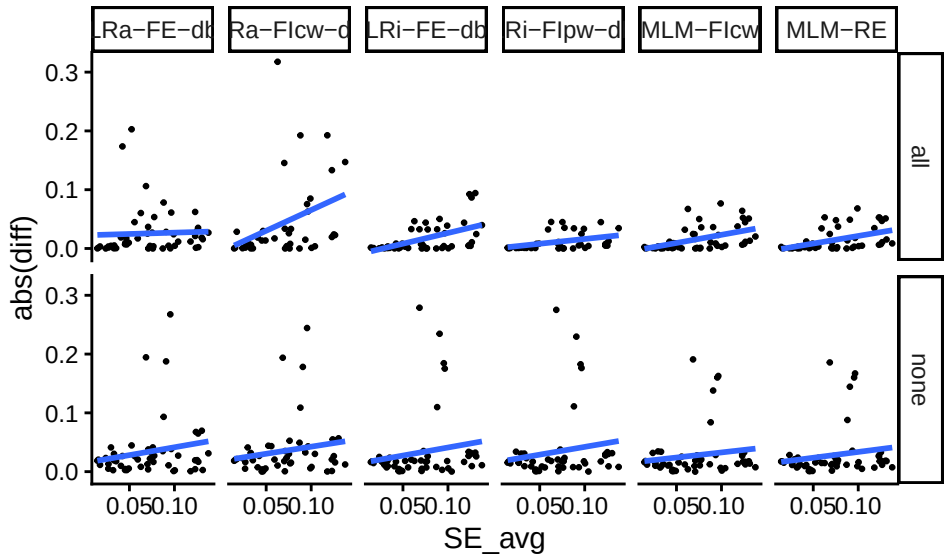
much, regardless of estimator.

We can summarize average differences by estimator and adjustment:

Estimator	Estimand	Covariates	n	avg_diff	sd_diff	Q10	Q90
LRa-FE-db	Cluster	all	50	0.03	0.04	0	0.06
LRa-FE-db	Cluster	none	50	0.03	0.05	0	0.07
LRa-FIcw-db	Cluster	all	50	0.04	0.07	0	0.15
LRa-FIcw-db	Cluster	none	50	0.04	0.05	0	0.06
MLM-FIcw	Cluster	all	50	0.02	0.02	0	0.05
MLM-FIcw	Cluster	none	50	0.03	0.04	0	0.04
MLM-RE	Cluster	all	50	0.01	0.02	0	0.05
MLM-RE	Cluster	none	50	0.03	0.04	0	0.04
LRi-FE-db	Person	all	50	0.02	0.02	0	0.04
LRi-FE-db	Person	none	50	0.03	0.06	0	0.04
LRi-FIpw-db	Person	all	50	0.01	0.01	0	0.03
LRi-FIpw-db	Person	none	50	0.03	0.06	0	0.06

We can see if the size of the change in point estimate is related to the overall uncertainty in the study:





These features do not seem to explain when covariate adjustment matters a lot.

Here are the studies where there was massive change (above 0.15) when adjusting for covariates for at least one estimator:

Covariates	UniqueID	cons_df	J	K	nbar	R2.1	R2.2	n_est	avg_diff
all	CARES_DC_EMIT_PTH	8	52	22	23.63	0.17	0.95	1	0.32
all	CLC_GRCCSS_POST	10	28	9	341.32	0.17	0.69	1	0.20
all	CARES_TR_SSRS_PTH	8	52	22	24.54	0.20	0.38	1	0.19
all	jg_crt_12	16	30	7	87.07	0.44	0.81	1	0.19
all	CLC_PCCRERCORE	10	28	9	588.04	0.07	0.23	1	0.17
none	jg_crt_7	16	30	7	46.20	0.41	0.87	6	0.22
none	jg_crt_8	16	30	7	44.50	0.32	0.93	4	0.21
none	jg_crt_11	16	30	7	86.27	0.32	0.74	6	0.20
none	jg_crt_2	30	83	7	26.19	0.32	0.61	4	0.17

`n_est` is the number of estimators where the adjustment was bigger than 0.15.